Inverse Spherical Dual Hemisphere Quantum Mechanics

Physical Unit Counting and Operations

NOS v18

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Abstract

This paper applies the inverse mathematics established in *Foundation of Inverse Mathematics* (NOS v17) to derive the complete system of physical units and measurements. Building on the alternative counting system where numbers represent partitions of an existing whole rather than abstract accumulations from zero, we demonstrate that physical quantities emerge as exact **partition states** and **conjugate completions** within the four-quadrant mechanics.

We rigorously separate two orthogonal concepts:

1. **Unit operation** — the quadrant assignment (Q1–Q4), determined by how many successive Q1 bases (linear partitions) are compounded.
2. **Unit configuration** — the recursive depth ($1/128^n$), the resolution scale of any measurement.

The conventional "multiplication" in formulas such as work = force × distance is revealed as **inverse partition closure** — the exact reconstruction of the unpartitioned whole through conjugate pair completion. The fine-structure constant $\alpha^{-1} \approx 137$ emerges from the same gearbox partition that produces $\pi_{\text{machine}} = 201/64$, read as difference rather than ratio.

Prerequisites: This paper assumes familiarity with the Foundation paper (NOS v17), particularly the partition equality $n \times \frac{1}{n} = 1$, the 512-bin/720° structure, and the 128-bin quadrant block.

Contents

1 Introduction: From Foundation to Physics

The *Foundation of Inverse Mathematics* (NOS v17) established an alternative counting system where:

- Numbers represent partitions of an existing whole, not accumulations from zero
- The closed whole is $\sqrt{1} = 1 = 512$ bins $= 720°$
- The universal partition equality $n \times \frac{1}{n} = 1$ holds exactly
- The fundamental unit $u = 45/32°$ emerges from the $\sqrt{2}$ compression
- The 128-bin quadrant block provides the counting base for all integers

This paper applies that foundation to physics. We show that physical units are not dimensionless labels attached to abstract numbers, but **mechanical states** of the inverse sphere partitioned through conjugate pair operations.

The Physics Application

Physical quantities are partition states of the closed whole.
The title "**Inverse Spherical Dual Hemisphere Quantum Mechanics**" now names the **physics** — how the mathematical foundation operates in physical reality through conjugate pairs Q1↔Q3 (inner engine) and Q2↔Q4 (outer hull).

2 Core Definitions for Physical Application

2.1 Recall from Foundation

From NOS v17, we have the following established structures:

Foundation Structures (from NOS v17)

Structure	**Value**	**Origin**
Closed whole	$\sqrt{1} = 1$	Starting identity
Total bins	512	$\sqrt{512} = 2^{9/2}$ from partition mechanics
Total angular extent	720°	$9 \times 10 \times 8$ partition demand
Fundamental unit	$u = 45/32°$	Compressed $\sqrt{2}$
Quadrant block	128 bins	512/4 quadrants
Depth ladder	$u_Q = 1/128^Q$	Recursive partition refinement

2.2 Physical Reference Assignment

To apply the inverse mathematics to physics, we assign a physical reference to the full sphere measure:

Physical Reference — The Meter

The coarsest Q1-type measure (full sphere span in length) defines the physical meter:

$$\boxed{1\,\text{meter} = 512\,u = 720^\circ = 1\text{ inverse sphere}}$$

This is not arbitrary — it assigns the complete partition structure to the fundamental unit of length. All other physical units derive from this single assignment through partition operations.

3 Unit Operation — Quadrant Type

The **type** of a physical unit is determined by which quadrant's operation defines its dimensional character — specifically, how many Q1 bases (linear partitions) are compounded.

Quadrant Unit Types — Physical Assignment

Quadrant	Root	Bases	Operation	Physical Example	Dimensional Signature
Q1	$\sqrt{1}$	1	Linear cut	Length (meter)	L
Q2	$\sqrt{2}$	2	Area fold / rate	Velocity (m/s)	L / T
Q3	$\sqrt{3}$	3	Volume bin / tension	Force (newton)	M · L / T^2
Q4	$\sqrt{4}$	4	Hypervolume closure	Potential (volt)	M · L^2 / (T^3· Q)

Mechanical Origin of Unit Types

Each higher quadrant compounds one additional linear base:

- **Q1 (1 base):** Pure linear partition — length, displacement, position
- **Q2 (2 bases):** Two linear bases combined — velocity, rate, frequency
- **Q3 (3 bases):** Three linear bases — force, tension, momentum flux
- **Q4 (4 bases):** Four linear bases — potential, energy density, closure

This exactly matches the dimensional progression $\sqrt{n} = n$D established in the Foundation.

4 Unit Configuration — Depth Ladder

The **scale** of any measurement is its position on the recursive inverse partition ladder. This is orthogonal to unit type — a quantity can be any type at any depth.

Depth Configuration Formula

$$\boxed{L_{Qn} = L_{\text{whole}} \times \frac{1}{128^n} = 512\,u \times 128^{-n} = 2^{9-7n}\,u}$$

where n is the depth level (1, 2, 3, 4, ...).

Depth Configuration Scales (relative to 1 meter)

Depth n	Configuration	Length in u	Exact power of 2	Scale
0	1 meter	$512\,u$	2^9	Macroscopic
1	1 meter / 128^1	$4\,u$	2^2	Centimeter scale
2	1 meter / 128^2	$1/32\,u$	2^{-5}	Micrometer scale
3	1 meter / 128^3	$1/4096\,u$	2^{-12}	Nanometer scale
4	1 meter / 128^4	$1/524288\,u$	2^{-19}	Sub-atomic scale

4.1 Reversibility — Universal Closure at Every Depth

The depth ladder is fully reversible because it is pure dyadic inverse partitioning. Every finer depth recovers the full closed whole when multiplied by the exact number of subunits:

Closure Verification at Each Depth

- Depth 1: $4\,u \times 128 = 512\,u$ ✓
- Depth 2: $(1/32)\,u \times 128^2 = (1/32) \times 16384 = 512\,u$ ✓
- Depth 3: $(1/4096)\,u \times 128^3 = (1/4096) \times 2{,}097{,}152 = 512\,u$ ✓
- Depth 4: $(1/524288)\,u \times 128^4 = (1/524288) \times 268{,}435{,}456 = 512\,u$ ✓

General proof:

$$2^{9-7n}\,u \times 128^n = 2^{9-7n} \times (2^7)^n\,u = 2^{9-7n+7n}\,u = 2^9\,u = 512\,u$$

The partition equality (regions × size each = whole) holds exactly at every depth.

4.2 Type and Depth Are Orthogonal

A physical quantity has both a **type** (quadrant) and a **depth** (scale). These are independent:

Example: Force at Different Depths

A newton (Q3 type) can exist at any depth:

- Newton at depth 0: N_0 = macroscopic force
- Newton at depth 2: $N_0 \times 1/128^2$ = microscopic force
- Newton at depth 4: $N_0 \times 1/128^4$ = sub-atomic force scale

Same type (Q3), different configuration (depth). The partition equality is preserved at all scales.

5 The Core Inversion: Multiplication as Conjugate Completion

This section contains the central physical insight of the inverse framework.

The Standard View

In standard physics:

$$\text{Work/energy} = \text{force} \times \text{distance}$$

This is interpreted as **accumulation** — force applied repeatedly over distance, building up from zero.

The Inverse View

In inverse physics, there is no external multiplication and no accumulation from void.
"Force × distance" is **conjugate pair completion** — the exact closure of the partitioned whole back to unity.
The "×" symbol is the same glyph, but its logical meaning is inverted:

- **Standard:** accumulate from zero → external product
- **Inverse:** partition completion → internal closure to one

Mechanical Sequence: Force × Distance → Energy

1. **Start:** Closed whole $\sqrt{1} = 1 = 512$ bins
2. **Partition to Q3:** $\sqrt{3}$ creates 3 regions (volumetric bins) — this is the tension state we call **force**
3. **Apply Q1 displacement:** One linear partition ($\sqrt{1}$ = distance) operates across the Q3 tension field
4. **Complete to Q4:** The Q1 displacement completes the fourth partition: $\sqrt{3} \times \sqrt{1} \to \sqrt{4} = 2$
5. **Closure:** The partition equality closes exactly:

$$\boxed{4 \times \frac{1}{4} = 1 \quad \Rightarrow \quad \text{energy} = \pm 1u \quad \text{(oriented whole)}}$$

Energy Is Not Created — It Is Reconstructed

Energy is not "created" by multiplying inputs from nothing.
Energy is the **reconstruction** of the unpartitioned whole through conjugate operation — one thing operating through itself.
This is why energy is conserved: you cannot create or destroy the whole. You can only partition it differently. Every "energy transformation" is a repartitioning that preserves the total closure to unity.

6 Derivation of the Fine-Structure Constant

The most stringent test for any foundational system is the derivation of dimensionless physical constants. The fine-structure constant α has an inverse measured (CODATA 2022) as:

$$\alpha^{-1} \approx 137.035999206$$

In standard physics, this is an independent empirical input with no known mechanical origin. In the inverse framework, α emerges from the **same partition event** that produces π_{machine}.

The Gearbox Wall — Single Source

At the critical Q1→Q2 transition (90° gearbox wall), the partition structure produces:

- **Flat radial lattice count:** exactly 64 bins
- **Curved arc count** (demanded by inverse closure): exactly 201 bins

These two numbers — 64 and 201 — encode **both** π and α^{-1} through different operations on the same partition state.

Dual Reading of the Gearbox Partition

The same partition state (64 flat, 201 curved) admits two orthogonal readings:
Ratio operation (division):

$$\pi_{\text{machine}} = \frac{\text{curved bins}}{\text{flat bins}} = \frac{201}{64} = 3.140625$$

Difference operation (subtraction):

$$\alpha^{-1}_{\text{raw}} = \text{curved bins} - \text{flat bins} = 201 - 64 = 137$$

The same two numbers, read by different operations, produce two fundamental constants.

Applying the Universal Slippage

The gearbox slippage — the echo of the original $\sqrt{2} \to 45/32$ compression — adjusts both readings proportionally:

$$f_{\text{slip}} = \frac{\pi_{\text{observed}}}{\pi_{\text{machine}}} = \frac{3.14159...}{3.140625} \approx 1.0003081086$$

Applying this factor to the raw electromagnetic register:

$$\alpha^{-1}_{\text{predicted}} = 137 \times f_{\text{slip}} \approx 137.04221088$$

This matches the observed value to within $\Delta \approx 0.0062$ (relative error $\sim 4.5 \times 10^{-5}$).

Significance

The integer backbone (137) is captured **exactly**. The leading decimal tail emerges from the **same** mechanical source as π.
The strength of the electromagnetic interaction is not a free parameter but the **subtractive residue** of the π-machine partition when read in offset mode.
This eliminates α from the list of unexplained fundamental constants, unifying circular aperture (π) and charge coupling (α) under one gearbox event.

7 Mathematical Validation

7.1 Closure at Every Depth

Demonstrated in Section 4.1 — exact for all n.

7.2 Unit Composition Preserves Quadrant Type

Composition Verification

- **Velocity:** length (Q1) / time (cycle = 512 u) → Q2 type ✓
- **Force:** mass (Q3 tension) × acceleration (Q2 / cycle) → Q3 type ✓
- **Energy:** force (Q3) × distance (Q1) → completes to Q4 hypervolume ✓

All compositions match the 1-2-3-4 base count exactly.

7.3 No Irrational Slippage Introduced

All scaling is exact powers of 2:

- Total bins: $512 = 2^9$
- Quadrant block: $128 = 2^7$
- Depth scaling: $128^n = 2^{7n}$

The only compression is the inherited $\sqrt{2} \to 45/32$ per bin — already mechanized in u and accounted for in the slippage factor.

7.4 Conservation Laws as Partition Closure

Why Energy Is Conserved

In the inverse framework, conservation is not an empirical observation requiring explanation — it is a **logical necessity**.

The whole cannot be created or destroyed. It can only be partitioned differently. Every physical process is a repartitioning that preserves closure to unity:

$$n \times \frac{1}{n} = 1 \quad \text{(always)}$$

"Conservation of energy" is simply the statement that the partition equality holds before and after any transformation.

8 Conclusion

Building on the Foundation of Inverse Mathematics (NOS v17), this paper has derived the complete system of physical units from inverse partitioning of the single closed whole.

Summary of Physical Unit System

Concept	Mechanical Origin
Unit operation (type)	Quadrant assignment (Q1–Q4) = number of Q1 bases compounded
Unit configuration (scale)	Depth ladder ($1/128^n$) = resolution of measurement
Physical reference	1 meter = 512 u = full sphere span
Multiplication	Conjugate completion, not accumulation
Energy	Reconstruction of unity through Q3 + Q1 → Q4 closure
Conservation	Logical necessity of partition equality
Fine-structure constant	Difference reading of gearbox partition (201 - 64 = 137)

The Central Insight

There is no "times" in the accumulative sense.
"Force × distance" is conjugate completion: Q3 tension + Q1 displacement → Q4 hypervolume closure → energy = reconstruction of unity.
Standard physics counts the shadows of inverse partition. This framework reveals the mechanics casting those shadows.

Series Context

This paper (NOS v18) applies the alternative counting system to physical units. The series continues with:

- Quantum mechanics as partition depth operations
- Electromagnetic phenomena from gearbox geometry
- Gravitational mechanics from compression echo distribution
- Cosmological applications of the 720° closure

All built on the single foundation: $\sqrt{1} = 1$ — the whole already IS.